

# The Readout Regime: A Normal Form for Final-Residual Control of Frozen Transformers — and Its Capacity Limits

Nathan Lee Peterson

Independent researcher

Correspondence: [orbitnate@gmail.com](mailto:orbitnate@gmail.com)

Code and result artifacts: <https://github.com/orbitnate/readout-regime>

## Abstract

Inference-time interventions on a frozen transformer — steering behavior, injecting facts, suppressing outputs — are not interchangeable: *where* an additive intervention acts splits them into two regimes of sharply different expressive power, and we give the exact theory of one. An intervention that writes into the **final residual stream** (the unembedding/readout space; installing a scaled row of the output-projection matrix `lm_head` is the exactly-characterizable case) induces on the next-token logits, for every input, the transform  $z \mapsto s(x) \cdot z + c(x)$ : a ranking-preserving scalar temperature  $s(x) > 0$  plus a re-ranking bias  $c(x)$  **confined to a fixed, low-dimensional set of directions chosen before any input is seen** (T1; verified against direct forward-pass computation to  $\approx 5 \times 10^{-6}$ ). The result that matters is a *structure* theorem (T2): the input selects only a *point* in that fixed set and a temperature, so the reachable re-ranking directions, over all inputs, have affine dimension at most the installed-slot count. **In plain terms: a readout install can re-weight and re-rank the options the model already has, but cannot synthesize a new answer direction or compute a hidden routing variable the addressing query does not already expose.** We prove the readout regime’s confinement and *cite* — not prove — evidence that the representation regime is not so confined; that direct measurement is the main open item. The corollaries, carefully bounded: a bounded readout install is **not a hard override** of a peaked prior, can only **tip** a decision the context has already scaffolded near-balanced, and **cannot compute** a hidden intermediate. Capacity has two faces, both empirical: across **key-disjoint** decisions installs compose bit-exactly at scale (tens of modules, thousands of facts,  $\Delta = 0$ ); **within one** decision the readout is winner-take-all ( $\approx 2$  targets co-winnable, against an output projection of entropy-effective rank  $\approx 918$ ). The control reading: *tip* a propensity in the readout regime — it is auditable, a removable bias — but place any hard guarantee in a deterministic override outside the model. The contribution is the boundary, stated precisely.

## 1 Introduction

A growing toolbox modifies a frozen language model at inference time rather than by fine-tuning: activation/representation steering, function vectors, knowledge editors, attention-injected memories, and logit-space biases. These methods are attractive precisely because they are cheap, removable, and leave the base weights untouched. But they are not interchangeable, and the literature has not drawn a clean line between what each *can* and *cannot* do.

This paper draws that line for the case where the intervention is **additive in the final residual stream**: a vector added to the hidden state at the current decoding position, whose effect reaches the unembedding through the model’s final normalization with no further nonlinear processing. The most exact instance — and our analytic anchor — is installing a scaled **row of the output-projection matrix** (`lm_head`), the “value = unembedding row of a target token” construction. That `lm_head` rows are special directions in this space is not new: the observation that “row  $i$  of `lm_head` is precisely the direction that produces high probability for token  $i$ ” appears in recent work on retrofitting memory to frozen models (Wind, 2026, §4.4.3; see §8). We do not claim that observation. We claim its consequences, and we prove them.

### Contributions.

1. **T1 — exact readout transform (known core; exact form).** A final-residual additive install induces, on the next-token logits, the transform  $\tilde{z}(x) = s(x) \cdot z(x) + c(x)$ , where under RMSNorm  $s(x) = r/r' > 0$  is a scalar temperature and  $c(x)$  is a combination of **fixed** directions  $u_t = U(\gamma \odot U[t])$ . The no-norm case is the exact additive Gram-column bias  $\tilde{z} = z + \alpha \cdot UU[t]$ ; the LayerNorm case folds in a fixed centering (Lemma 4). We verify the closed form numerically against direct forward-pass computation to max-diff  $4.77 \times 10^{-6}$  (Mistral-7B) and  $7.15 \times 10^{-6}$  (Qwen2.5-7B).
2. **T2 — cone-confinement structure theorem (the contribution).** Across **all** inputs, a readout install’s re-ranking term  $c(x)$  lies in a single **fixed** set  $C$  of affine dimension  $\leq m$  (the slot count) — the convex cone generated by the installed (possibly signed) generators — chosen before any input. The input selects only a *point* in  $C$  (via key-addressed weights) and a temperature. Hence the reachable set of re-ranking directions, over all inputs, has rank  $\leq m$ , and a readout install **cannot synthesize a re-ranking direction outside the pre-installed set** (defined precisely in §4.3). This holds *even with* input-conditional, key-addressed attention over an arbitrary stored set, and uses only that the stored values are fixed before input and the weights are non-negative — *not* the `lm_head` identity (which is what makes T1 exact, §2 remark). The unconditional,  $m$ -independent content is **fixedness** (no input creates a re-ranking direction outside the  $m$  pre-stored generators); the *rank  $\leq m$  number* bites only when  $m$  is small relative to the answer cardinality — it is vacuous once  $m$  approaches the output projection’s  $\approx 918$  effective rank. (T2(a)–(c) follow from Lemmas 1/3/4; the load-bearing novelty is this fixedness/capacity statement and its measurable form, Corollary T2’, not the per-input algebra.)
3. **Corollaries (operational, carefully bounded).** T2 implies a readout install (C1) gives **no hard override** of a peaked prior — a bounded install flips a decision only when its fixed-cone correction beats the scaled base gap; (C2) can *tip* a context-scaffolded near-balanced decision; and (C3) cannot *compute or synthesize* a hidden routing variable or a new answer direction — it can exploit a hidden intermediate only insofar as the addressing query already exposes it. Each is a bounded consequence, not a sweeping impossibility.
4. **Capacity observations (empirical, not a theorem).** Across **key-disjoint** decisions, installs are additive and non-interfering at scale (tens of modules holding thousands of facts, bit-exact  $\Delta = 0$ ). Within a single decision,  $\approx 2$  targets are cleanly co-winnable, against an output-projection whose entropy effective rank we measure at  $\approx 918$  (stable rank  $\approx 10$ ). We label these as evidence, not a proved bound.
5. **An empirical map**, split by install site into (A) final-residual installs that behave exactly as T1 predicts and (B) late-layer installs where the same boundary pattern is empirically visible — bias-only tasks succeed, computation/chaining tasks hit the wall.
6. **A control reading.** The same limit that prevents *computed or hidden-routed* knowledge-injection via the readout regime (single-token recall on a flat prior still works — §7) tells you

which control jobs belong there (propensity tips, auditable as a removable bias) and which require a deterministic override outside the model (hard gates).

**Known vs. new.** The closed-form equivalence (T1) is essentially known — adding an `lm_head` row biases its token (Wind, 2026; §8). Our contribution is its *structure*: the cone-confinement theorem (T2), the falsification-probe family (§6), and the capacity observations (§5). One scope limit, stated up front: we prove cone-confinement of the readout regime and cite empirical evidence that the representation regime is *not* so confined, but we do **not** claim a formal containment lattice between the two — the distinction is structural, not a proven  $\subseteq/\subsetneq$  (§4.4).

### Claim status at a glance.

| Claim                                                     | Status                                                             |
|-----------------------------------------------------------|--------------------------------------------------------------------|
| T1 — readout transform closed form                        | <b>proved</b> ; numerically checked ( $\approx 5 \times 10^{-6}$ ) |
| T2 / T2' — cone-confinement; affine rank $\leq m$         | <b>proved</b>                                                      |
| Capacity within one decision ( $\approx 2$ co-winnable)   | empirical observation                                              |
| Additivity across key-disjoint decisions ( $\Delta = 0$ ) | empirical observation                                              |
| Representation regime is <i>not</i> cone-confined         | <b>cited</b> (Nadaf, 2026); not directly measured here             |

## 2 Setup and notation

Fix a frozen transformer language model and a single decoding position. For input context  $x$ :

- $h(x) \in \mathbb{R}^d$  — the residual-stream state at the **final** layer and current position, immediately **before** the final normalization;
- $N : \mathbb{R}^d \rightarrow \mathbb{R}^d$  — the final normalization (identity; RMSNorm with gain  $\gamma \in \mathbb{R}^d$  and  $\varepsilon > 0$ ; or LayerNorm);
- $U \in \mathbb{R}^{|V| \times d}$  — the unembedding / output-projection matrix (`lm_head.weight`), rows  $U[j]$ ;
- $z(x) = U \cdot N(h(x)) \in \mathbb{R}^{|V|}$  — base next-token logits;  $p(x) = \text{softmax}(z(x))$ .

Write  $G := UU^\top$  (Gram matrix of unembedding rows),  $g_t := UU[t] = G[:, t]$ , and the  $\gamma$ -**weighted** column  $u_t := U(\gamma \odot U[t])$  (so  $u_t = g_t$  when  $\gamma = 1$ ).

### Notation summary.

| Symbol                                    | Meaning                                                                                 |
|-------------------------------------------|-----------------------------------------------------------------------------------------|
| $h(x) \in \mathbb{R}^d$                   | final-layer residual at the current position, before the final norm                     |
| $N, r(h), \gamma$                         | final normalization; RMSNorm denominator $\sqrt{\text{mean}(h^2) + \varepsilon}$ ; gain |
| $U \in \mathbb{R}^{ V  \times d}, U[t]$   | unembedding ( <code>lm_head.weight</code> ); row of target token $t$                    |
| $z(x) = U \cdot N(h(x))$                  | base next-token logits; $p(x) = \text{softmax}(z)$                                      |
| $m, (t_i, K_i, \alpha_i)$                 | slot count; slot $i$ 's target token, key, and fixed scale                              |
| $a_i(x) \geq 0$                           | input-addressed (non-negative) weight on slot $i$                                       |
| $\Delta(x)$                               | residual perturbation added before $N$ , eq. (1)                                        |
| $g_t = UU[t], u_t = U(\gamma \odot U[t])$ | Gram column; $\gamma$ -weighted column (the fixed bias directions)                      |
| $s(x) = r(h)/r(\tilde{h})$                | structural temperature ( $\tilde{h} = h + \Delta$ )                                     |
| $c(x) \in C$                              | re-ranking term; $C$ = fixed re-ranking set, $\dim \text{span}(C) \leq m$               |
| $d(x) = \tilde{z}(x) - s(x)z(x)$          | re-ranking residual (Corollary T2')                                                     |

**The readout-regime install.** A readout install is a finite set of *slots*

$$S = \{(t_1, K_1, \alpha_1), \dots, (t_m, K_m, \alpha_m)\},$$

where slot  $i$  stores value  $V_i = \alpha_i \cdot U[t_i]$  (a scaled **lm\_head** row; the scale  $\alpha_i \in \mathbb{R}$  is fixed before any input) and key  $K_i$ . At decode, a query  $q(x)$  (an input-addressing read of the current residual stream) produces **non-negative** weights  $a_i(x) \geq 0$  over the slots (e.g. a softmax over  $\langle q(x), K_i \rangle$ , optionally with an attention sink so  $\sum_i a_i \leq 1$ ), and the install adds to the final residual, before  $N$ :

$$\Delta(x) = \sum_{i=1}^m a_i(x) \cdot V_i = \sum_i a_i(x) \cdot \alpha_i \cdot U[t_i] \in \mathbb{R}^d, \quad (1)$$

giving  $\tilde{h}(x) = h(x) + \Delta(x)$  and installed logits  $\tilde{z}(x) = U \cdot N(\tilde{h}(x))$ . The simplest case — one always-on slot,  $m = 1$ ,  $a_1 \equiv 1$  — recovers the bare install  $\Delta = \alpha \cdot U[t]$ .

**Scope of the abstraction.** Eq. (1) is the *final-residual, direct-addition* object: a vector added straight to the residual that reaches the unembedding through  $N$  only.

- *We deliberately do not model interventions injected earlier in the stack* (at an MLP down-projection or attention output of a non-final layer), whose effect passes through the remaining nonlinear layers before readout. Those are *representation-regime* interventions and T2 does not govern them. (This is why §7 is split: Tier A installs are literally eq. (1); Tier B late-stack injections are reported as consistent-with, not as tests of the object.)
- *Direct addition vs. attention-head addition* ( $W_O$ ). Eq. (1) adds the stored value to the residual directly. A real attention head would post-multiply by the output projection  $W_O$ , so the unembedding image of a stored **lm\_head** row would be  $U \cdot W_O \cdot U[t]$ , not the clean Gram column  $g_t = U \cdot U[t]$ . The final-residual embodiment we analyze (a “pure-geometric” install, residual  $\leftarrow$  residual +  $c \cdot A$  at the final-norm pre-hook) is a **direct addition**, so Lemma 1’s exact Gram-column form is faithful to it. If a memory is instead realized as a genuine attention head, the generator becomes  $\text{cone}\{UW_O\alpha_i U[t_i]\}$  — **still a fixed cone**, so T2(c) survives unchanged; only Lemma 1’s *exact* Gram-column identity and the §6 probes (which isolate the bare **lm\_head** row) require the direct-addition reading.
- *The query need not be linear.* We do not assume  $q(x)$  is linear in  $x$ ; T2 uses only that the resulting weights  $a_i(x)$  are non-negative and that the keys/values are fixed before input. This *strengthens* the result (it is robust to an arbitrarily nonlinear input-addressing read).

**The defining invariant.** The stored directions  $\{U[t_i]\}$ , the scales  $\{\alpha_i\}$ , and the keys  $\{K_i\}$  are **fixed before any input is seen**; the only input-dependent quantities are the non-negative weights  $a_i(x)$ . Everything below follows from *fixed-pre-input values + non-negative weights*.

(Remark — *generality vs. attribution.*) The cone argument below uses only that the stored directions are fixed before input and the weights are non-negative; it does not use that the values are specifically **lm\_head** rows. So T2 is broader than the **lm\_head** construction. The **lm\_head** identity is what makes T1 **exact** (the bias direction is  $u_t$ , a known unembedding column) and what the §6 probes isolate; it is not required for the impossibility. The only way to escape T2 is to make the stored value itself an input-conditional nonlinear function of  $x$  (a genuine  $W_V \cdot N(\text{context})$  value) — at which point the intervention is a *representation-regime* computation, not a readout install. That closes the escape.)

### 3 T1 — The readout transform (closed form)

**Lemma 1** (no-norm, exact). *If  $N = \text{Id}$ , then for every input  $x$ ,  $\tilde{z}(x) = z(x) + \sum_i a_i(x) \cdot \alpha_i \cdot g_{t_i}$ , with  $g_t = UU[t] = G[:, t]$ . In the always-on single-slot case this is  $\tilde{z} = z + \alpha \cdot g_t$ , a **fixed additive logit bias**  $b = \alpha \cdot G[:, t]$ : token  $t$  gains  $\alpha \|U[t]\|^2$ ; token  $j$  gains  $\alpha \langle U[j], U[t] \rangle$ .*

*Proof.*  $\tilde{z} = U(h + \Delta) = z + U\Delta = z + \sum_i a_i \alpha_i UU[t_i]$ .  $\square$

**Lemma 2** (reachable re-ranking set). *For any  $x$ ,  $m$ , weights  $a_i(x) \geq 0$ , and fixed scales  $\alpha_i$ , the induced logit perturbation  $U\Delta(x)$  lies in the **fixed set***

$$C := \{ \sum_i a_i \alpha_i g_{t_i} : a_i \geq 0 \} \subseteq \text{span}\{g_{t_1}, \dots, g_{t_m}\} \subseteq \text{col}(U).$$

*If all  $\alpha_i \geq 0$ ,  $C = \text{cone}\{g_{t_i}\}$  (a convex cone). If a scale  $\alpha_i$  is negative, that slot contributes the signed generator  $\alpha_i g_{t_i}$ , and  $C$  is the convex cone over the signed generators actually installed —  $\text{cone}\{\alpha_i g_{t_i}\}$  — which equals the full subspace  $\text{span}\{g_{t_i}\}$  iff the signed generators positively span it (installing each with both signs is sufficient but not necessary). In all cases  $\dim \text{span}(C) \leq m$ , and  $C$  is determined entirely by  $(U, \{t_i\}, \{\alpha_i\})$  — chosen **before any input**.*

*Proof.* Each  $g_{t_i} \in \text{col}(U)$ ; a non-negative combination of  $\{\alpha_i g_{t_i}\}$  is in the stated set; its span has dimension  $\leq m$ .  $\square$

**Lemma 3** (RMSNORM — the temperature term). *Let  $N(h) = \gamma \odot h/r(h)$ ,  $r(h) = \sqrt{\text{mean}(h^2) + \varepsilon}$ . Then for every input  $x$ ,*

$$\begin{aligned} \tilde{z}(x) &= s(x) \cdot z(x) + \sum_i b_i(x) \cdot u_{t_i}, \\ s(x) &= r(h(x))/r(\tilde{h}(x)) > 0, \quad b_i(x) = a_i(x) \cdot \alpha_i / r(\tilde{h}(x)), \quad u_t = U(\gamma \odot U[t]). \end{aligned} \tag{2}$$

*Equivalently, the induced bias  $\tilde{z} - z$  is  $(\alpha/r') \cdot u_t + ((r - r')/(r \cdot r')) \cdot U(\gamma \odot h)$  for the single-slot always-on case. We verify this identity numerically against direct forward-pass computation to **max-diff**  $4.77 \times 10^{-6}$  (**Mistral-7B-Instruct-v0.3**) and  $7.15 \times 10^{-6}$  (**Qwen2.5-7B**), with the additive term carrying a median 96.1% (Mistral; 96.83% on Qwen) of  $\|\tilde{z} - z\|$  and the rescale term argmax-invariant.*

*Proof.*  $\tilde{z} = U(\gamma \odot (h + \Delta))/r' = (r/r') \cdot [U(\gamma \odot h)/r] + (1/r') \cdot U(\gamma \odot \Delta) = s \cdot z + \sum_i (a_i \alpha_i / r') \cdot u_{t_i}$ ,  $r' := r(\tilde{h})$ . For the single-slot always-on case,  $s \cdot z - z = ((r - r')/r') \cdot [U(\gamma \odot h)/r] = ((r - r')/(r \cdot r')) \cdot U(\gamma \odot h)$ , and the additive term is  $(\alpha/r') \cdot u_t$ . (The  $\varepsilon$  in  $r, r'$  is retained; it perturbs  $s$  by  $O(\varepsilon/\|h\|^2)$  and affects no ordering result.)  $\square$

**Lemma 4** (LayerNorm — centering folds in). *Let  $N(h) = \gamma \odot (Ph)/\sigma(h)$  (biasless LayerNorm; the additive- $\beta$  case is in the remark below) with the centering projection  $P = I - (1/d)\mathbf{1}\mathbf{1}^\top$  and  $\sigma(h) = \sqrt{\text{mean}((Ph)^2) + \varepsilon}$ . Then  $\tilde{z}(x) = (\sigma/\sigma') \cdot z(x) + (1/\sigma') \cdot \sum_i a_i \alpha_i \cdot \tilde{u}_{t_i}$ , where  $\tilde{u}_t := U(\gamma \odot PU[t])$  is a **fixed direction** and  $\sigma' := \sigma(\tilde{h})$ . The temperature  $\sigma/\sigma' > 0$ , and the re-ranking set  $\tilde{C} = \{\sum_i a_i \alpha_i \tilde{u}_{t_i}\}$  has the same fixed- $\leq m$ -dimensional structure as Lemma 2.*

*Proof.*  $P$  and  $\gamma \odot (\cdot)$  are fixed linear maps;  $z = (U \text{diag}(\gamma)P)h/\sigma$ . Adding  $\Delta$ :  $\tilde{z} = (U \text{diag}(\gamma)P)(h + \Delta)/\sigma' = (\sigma/\sigma')z + (1/\sigma')U \text{diag}(\gamma)P\Delta$ , and  $U \text{diag}(\gamma)PU[t_i] = \tilde{u}_{t_i}$  is fixed.  $\square$

*Remark (final LayerNorm bias  $\beta$ ).* With  $N(h) = \gamma \odot (Ph)/\sigma + \beta$ , the logits carry an extra term  $U\beta$ ; the re-ranking residual then gains  $(1 - \sigma/\sigma') \cdot U\beta$  — **one additional fixed direction**  $U\beta$ , input-modulated only by the scalar temperature mismatch. The set  $C$  grows to  $\leq m + 1$  fixed directions, still a fixed finite direction set, so the cone-confinement of T2 is unchanged. Most modern decoders (RMSNORM-family, biasless LayerNorm) have no  $\beta$ ; we note this only for completeness.

So under any of  $\{\text{Id}, \text{RMSNORM}, \text{LayerNorm}\}$ , a readout install is **exactly** a positive per-input rescale of the base logits plus a combination of **fixed** directions. The headline survives normalization: **readout install = fixed-direction-set bias + scalar temperature**.

**Relation to prior work.** The no-norm core (Lemma 1 — adding an `lm_head` row biases its token) is the observation Wind (2026, §4.4.3) and informal “logit-gap” notes already make. T1’s job is to state it *exactly* and expose the structure (the fixed direction set, the temperature) that T2 needs.

## 4 T2 — Cone-confinement structure theorem (the contribution)

### 4.1 The per-input action, stated precisely

By Lemmas 1/3/4, a readout install acts on every input as

$$\tilde{z}(x) = s(x) \cdot z(x) + c(x), \quad s(x) > 0, \quad c(x) \in C, \quad (3)$$

where  $C$  is the **fixed** re-ranking set of Lemma 2 ( $\dim \text{span}(C) \leq m$ , chosen before any input) and  $s(x)$  is a scalar temperature.

**What the temperature does, stated correctly.** For any  $s > 0$ ,  $s \cdot z$  preserves the *entire ranking* of  $z$  — a pure temperature changes confidence (entropy), never order. It does **not** follow that “all re-ranking lives in  $c$  independent of  $s$ ”: the *realized* re-ranking is determined **jointly**, because  $c$  competes against the scaled base logits  $s \cdot z$  (large  $s$  lets the base dominate; small  $s$  lets  $c$  dominate). Two inputs with the same cone-point  $c$  but different  $s$  can therefore differ in which pairs flip. The structure claim (§4.2) is about the *available re-ranking directions* — those, and only those, in the fixed  $C$  — and is unaffected by this interaction.

### 4.2 The structure theorem

**Theorem T2** (cone-confinement). *Let an intervention be a readout install (§2; values fixed before input, weights non-negative, direct final-residual addition). Then for every input  $x$ , eq. (3) holds with a single fixed  $C$  ( $\dim \text{span}(C) \leq m$ ) and  $s(x) > 0$ . Consequently:*

(a) **Fixed-bias core (no-norm, always-on).** *The intervention applies the same additive bias  $b \in C$  to every input:  $\tilde{z}(x) = z(x) + b$ , so  $\tilde{p}(x) = \Psi_b(p(x))$  with  $\Psi_b$  **independent of  $x$** . Inputs the base model maps to equal output distributions are mapped identically.*

(b) **Pairwise-gap structure.** *For any token pair  $(j, k)$ ,  $\tilde{z}_j(x) - \tilde{z}_k(x) = s(x) \cdot [z_j(x) - z_k(x)] + (c_j(x) - c_k(x))$ , with  $c(x) \in C$ . In the always-on no-norm case  $s \equiv 1$  and  $(c_j - c_k)$  is a **constant**, so the post-install ranking of  $(j, k)$  is a fixed threshold of the base gap. In the general normed/attention case the correction varies with  $x$ , but **only through a point of the fixed  $C$  and the scalar  $s$** .*

(c) **Cone confinement (general).** *For arbitrary  $N \in \{\text{Id}, \text{RMSNORM}, \text{LayerNorm}\}$ ,  $m$ , and (possibly nonlinear) input-addressing weights  $a_i(x) \geq 0$ , the re-ranking term  $c(x)$  lies, for **every***

input, in the **fixed** set  $C$  ( $\dim \text{span}(C) \leq m$ ) chosen before any input. The input selects only a point in  $C$  and a temperature  $s(x) > 0$ . An attention sink only makes  $\sum_i a_i \leq 1$ , shrinking but never leaving  $C$ . (For a pure softmax addressor with no sink,  $\sum_i a_i = 1$ , so the reachable set is the convex-hull slice  $\text{conv}\{\alpha_i g_{t_i}\} \subseteq C$ ;  $C$  is a valid over-approximation and  $\dim \text{span}(C) \leq m$  is unaffected.)

*Proof.* Lemma 1 gives (a) and the  $s \equiv 1$  part of (b). Lemmas 3/4 give eq. (3) and hence general (b) and (c): the additive term is  $\sum_i b_i(x) u_{t_i}$  (resp.  $\tilde{u}$ ), a point of  $C$ ;  $s(x) > 0$  preserves the ranking of  $z(x)$ .  $\square$

**Corollary T2'** (capacity of input-conditional re-ranking). Define the **re-ranking residual** using the install's **structural** temperature  $s(x) = r(h(x))/r(\tilde{h}(x))$  (Lemma 3):  $d(x) := \tilde{z}(x) - s(x) \cdot z(x)$ . For a readout install,  $d(x) = c(x) \in C$  for all  $x$ , so the affine dimension of  $\{d(x) : x \in \text{inputs}\}$  is **at most**  $m$ . Any intervention whose re-ranking residual — measured at this **structural** temperature — spans affine dimension  $> m$  over the same input set is **not** realizable as an  $m$ -slot readout install. (We use the structural  $s(x)$ , not a least-squares best fit: a best-fit scalar absorbs part of  $c(x)$  and would inflate the apparent rank, so the exact  $d(x) = c(x)$  identity holds only at  $s = r/r'$ . This  $s(x)$  is computed from the install's own pre/post-norm ratio and has **no canonical analogue** for an arbitrary representation-regime intervention, so the measure is readout-anchored, not fully regime-neutral — see §4.3.)

### 4.3 Why this is a bound, not a tautology

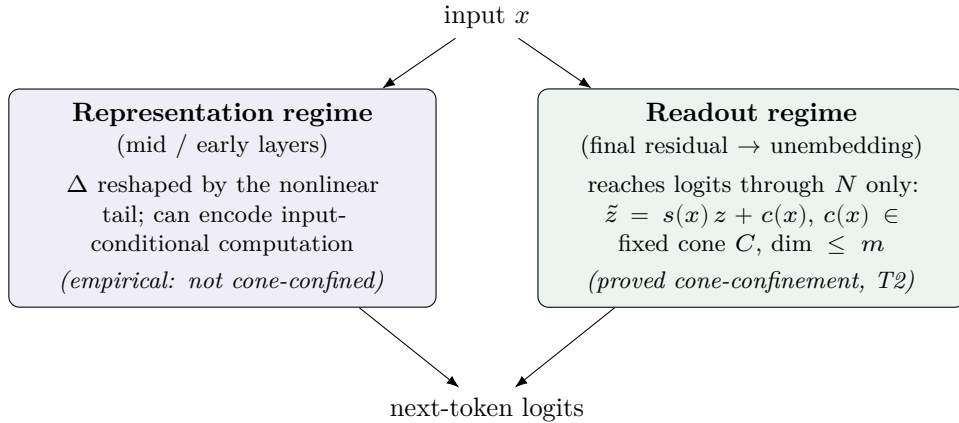
A reviewer's natural worry: if we *defined* "input-conditional computation" as exactly "a re-ranking not of the readout form," then "readout installs can't do it" would be true by construction. We avoid that. T2 is stated as a **structure theorem** (the reachable re-ranking set is a fixed  $\leq m$ -dim cone), and the substance lives in Corollary T2', a **capacity bound** measured by a quantity — the affine rank of the re-ranking residual  $d(x)$  — that names no mechanism. (One honest caveat:  $d(x)$  subtracts the install's *structural* temperature  $r/r'$ , which has no canonical analogue for a representation intervention, so the measure is readout-anchored rather than fully regime-neutral; the non-tautological content is that this rank is capped at  $m$  for the readout install no matter how nonlinear the addressor is.)

**Definition** (input-conditional re-ranking, regime-independent). An intervention performs *input-conditional re-ranking of order  $k$*  on an input set  $X$  if, after removing the best-fit positive scalar multiple of the base logits for each input, the residual re-ranking vectors  $\{d(x)\}$  span affine dimension  $k$ , and the map from input to residual direction depends on an intermediate variable not fixed as part of the intervention.

This definition refers only to the residual rank and the dependence on a non-fixed intermediate; it names no mechanism. (Note the two temperatures: this regime-*neutral* order uses the per-input **best-fit** scalar, because an arbitrary intervention has no canonical temperature; the readout **bound** of Corollary T2' is stated at the install's **structural**  $r/r'$ . The two coincide only up to the temperature mismatch — under best-fit, a readout install's *apparent* order can exceed  $m$  because the mismatch leaks the cone's own components into  $d(x)$ . So the clean  $\leq m$  statement is the structural one; the best-fit definition gives a weaker, fully regime-neutral comparison.) The readout-specific limitation is then a separate statement: a readout install can realize such variation **only by selecting among its pre-stored directions** — it cannot create new directions, and successful routing to a hidden bridge requires the addressor to already expose that bridge.

A 2-hop task whose answer token depends on a hidden bridge taking  $B$  values needs the re-ranking direction to access  $B$  distinct answer directions *selected by the bridge value*. T2 does **not** constrain what the addressor  $q(x)$  has computed —  $q(x)$  may be arbitrary and nonlinear; T2 constrains only what happens *after* the addressor reduces the input to non-negative weights over the fixed stored values. So the **unconditional** bound is: a readout install **cannot synthesize a new answer direction** outside the installed  $m$  — even at  $m \geq B$ , no input produces a re-ranking direction the install did not pre-store. Whether it can *route* to the right stored direction is **conditional** on the addressor: it routes correctly iff  $q(x)$  already exposes the bridge (the final residual linearly encodes it), and it **cannot compute** a bridge the residual does not already expose. The non-tautological content of T2 is the *direction-creation* limit ( $\text{rank} \leq m$ ), which holds regardless of the addressor; the routing failure on any given surface is the separate, empirical question of whether that surface’s residual exposes the bridge (it does not, on the surfaces of §7).

#### 4.4 The two regimes: a structural distinction, not a containment lattice



**Figure 1:** *Where the intervention acts decides what it can do.* A representation-regime install (left) is reshaped by the nonlinear tail and can encode input-conditional computation that the logit lens cannot decode (Nadaf, 2026). A readout install (right) reaches the logits through the final norm only, so by T1/T2 its effect is a scalar temperature plus a re-ranking term confined to a fixed  $\leq m$ -dimensional cone chosen before any input. We prove the right-hand confinement; the left-hand non-confinement is cited as empirical.

We **do not** claim a formal lattice  $G_{\text{read}} \subseteq G_{\text{repr}}$  between the behaviors realizable by readout installs and by representation-regime (mid-layer) interventions. A general containment in either direction is not proven here: the “easy” direction  $G_{\text{read}} \subseteq G_{\text{repr}}$  is not obviously true (a fixed mid-layer perturbation pushed through the nonlinear tail need not reproduce an arbitrary final-residual cone bias), so the two classes may be **incomparable**, not nested. What we state is the **provable asymmetry**:

- **Readout regime (proved):** re-ranking is confined to a fixed  $\leq m$ -dim cone chosen before any input (T2); the re-ranking residual has affine rank  $\leq m$  (T2’).
- **Representation regime (empirical, not proved here):** existing function-vector results provide empirical evidence for mid/early-layer steering that is **not well described as a fixed answer-direction bias under the logit lens** — it steers even where the logit lens cannot decode the answer at any layer (Nadaf, 2026). We do not compute  $d(x)$  for those interventions, so we do not claim their residual rank exceeds a comparable readout install’s; the formal residual-rank comparison is future work.



*Illustrative witness (sketch, explicitly not a theorem).* A single mid-layer additive steer  $h_\ell \mapsto h_\ell + \Delta$  induces  $\tilde{z}(x) = U \cdot N(F_{\ell \rightarrow L}(h_\ell(x) + \Delta))$  with  $F_{\ell \rightarrow L}$  nonlinear; on a task where a hidden bit computed by  $F_{\ell \rightarrow L}$  gates which of two unembedding-incomparable directions the re-ranking takes,  $\{d(x)\}$  has affine rank  $> 1$  with input-driven selection — order-2 input-conditional re-ranking, which no  $m = 1$  readout install achieves. We offer this as an existence illustration consistent with Nadaf (2026); a model-independent separation (a concrete witness *and* the inclusion direction) is left open (§10).

*Remark (no  $\text{col}(U)$  sleight-of-hand).* Every logit vector from any intervention lies in  $\text{col}(U)$  trivially (it is  $U$  times something), so the separation is **not** “readout stays in  $\text{col}(U)$ , representation leaves it.” The sharp invariant is the **fixed cone**: a readout install’s re-ranking is a combination of a *pre-fixed*  $\leq m$ -dim direction set, identical across inputs. The representation-side “not cone-confined” half is, in this paper, an **empirical** fact (Nadaf, 2026), not a proof step — and “pre-fixed” means the installed slot set and keys are frozen *before the input*, explicitly excluding context-time / retrieval-augmented slot selection (where the stored set is chosen per query).

## 4.5 Corollaries

Three operational corollaries follow, each matched to evidence in §7 (tier stated there).

- **C1 — no hard override under bounded correction.** By T2(b), a pair (target, prior) flips only when  $c_{\text{target}}(x) - c_{\text{prior}}(x) > s(x) \cdot [z_{\text{prior}}(x) - z_{\text{target}}(x)]$ . For any **bounded** install (drift-limited per its calibration, so it stays multi-fact-safe), the left side is bounded by the cone geometry, so there exist base gaps large enough that no flip is possible: bounded readout installs are **not hard overrides**, even with the install direction exactly correct. (Some peaked priors *can* be flipped — the claim is the absence of a *guarantee* under bounded correction, not that every peaked prior survives. An *unbounded* install can force a target as argmax — Tier-B any-token, +20 nats — but only by corrupting every other pairwise gap: a blunt overwrite, not a controlled override.) *Evidence (Tier B, §7):* injecting the oracle-correct value (`V == lm_head[target]`,  $\cos \approx 1.0$ ) at the late residual, scaled  $\times 1 \dots \times 8$ , moves the target logit by only  $+0.016 \dots +0.136$  and the model stays on its peaked prior (8/32). The failure is structural, not a wrong-direction problem.
- **C2 — can only tip a context-scaffolded near-balanced decision.** When in-context structure flattens the base gap across the candidate set, a small bias from  $C$  selects the target — and the composition is performed by the **visible context, not by the injected module**, consistent with T2. *Evidence (Tier B, §7):* a composition surface with a **visible** runtime operand reaches 272/272 at the **oracle / exact-captured-per-m delta** — the *upper bound* (controls: no-module 0.2684, wrong-delta 0.0551, so the lift is real and bridge-independent). But the **learned hidden route on the same surface reaches only 189/272 = 0.695 and fails its pre-registered gate**, and that residual is itself K=4-codebook regression error, *not* module recovery — so C2’s support is the in-context scaffolding plus the controls, **not** any claim that the module *recovered* the operand. A separate residual-injection arm reaches 32/32 **only when the bridge→status table is visible in context**, with the no-bridge floor at 7/32.
- **C3 — cannot compute a hidden intermediate (conditional).** A readout install cannot create a new answer direction or compute a routing variable (T2/T2’); it can route to a hidden bridge **only if** the addressing query already exposes it. So multi-hop chaining fails whenever the bridge is not linearly available to the addressor — an empirical property of the surface, not a T2 consequence alone (see §4.3). *Evidence (Tier B, §7), across three families:* installing the captured first-hop delta and forcing a second decode fails to chain — on **Llama-3.1-8B**

the strongest single-pass multi-hop cell is **0.183 vs a 0.40 bar** (n=372 MQuAKE-CF chains; McNemar canon-vs-control  $p = 2.15 \times 10^{-20}$ ), with the honest caveat that this surface installs at the *subject* positions and Phase-2 re-encodes the bridge, so it is a **locus-confounded near-miss**, not a pure final-residual wall; a **Llama→Mistral cross-host** install-then-continue gate reaches **0.120 vs a 0.70 bar** (n=50). A mid-layer “read” — on a small *from-scratch CCA substrate* (toy; artifacts off-repo) — is bias-equivalent (cosine 0.9918) with below-chance composition ( $2/16 < 0.25$ ) and payload-swap 0/3. A directly-built *hidden* second-hop surface failed its own positive control (8/32 at base rate) and is reported in §7 as a failed construction, not as a clean wall demonstration.

## 5 Capacity of the readout regime (empirical observations)

Co-installing  $\{(t_i, K_i, \alpha_i)\}$  gives, at a decision, the combined re-ranking term  $\sum_i b_i(x) u_{t_i}$ . Two *distinct* geometries govern capacity and must not be conflated:

- **Across decisions — key-disjointness.** When at most one module is addressed per decision ( $a_i \approx 0$  for non-matching modules), installs are **additive and non-interfering**. Empirically, co-loaded single-fact modules give **bit-exact** output ( $\Delta = 0$ ) and idempotent load/unload with `App1=0` against the unmodified model: 43 real-Wikipedia knowledge bases holding ~2,580 facts, per-KB co-load  $\Delta = +0.00$  pp at the 95th percentile, plus an N=100 synthetic co-load. The cross-decision capacity is, for practical purposes, unbounded *given key-disjointness* — this is a property of the keys, not a theorem about the values. (We say “tens of co-loaded modules holding thousands of facts,” not “thousands of modules,” to match the measured records.)
- **Within a decision — value-cone degeneracy.** Softmax is winner-take-all and the `lm_head`-row Gram is far from orthogonal, so the number of independently *co-winnable* targets at one decision is small — empirically  $\approx 2$ . (Source: a K-target packet probe — at K=2, **at least one of the two installed targets wins the argmax** on all three architectures (Llama 14/15, Mistral 15/15, Qwen 11/15) against a **substrate-clean random control** (zero hits across 1,600 random control packets); but the **strict both-win rate is only 3/15 (Llama) / 2/15 (Mistral) / 1/15 (Qwen)** — the readout is **winner-take-all** (K-way competition), *not* additive co-installation, and  $K \geq 3$  collapses. Corroborated by an in-context K-readability degradation 93.75% (K=4)  $\rightarrow$  64% (K=6)  $\rightarrow$  53% (K=8). We report  $\approx 2$  as the clean single-decision argmax ceiling.)

**Reconciling the two capacity figures.** The single-decision argmax ceiling ( $\approx 2$ ) and the **entropy-effective rank of the output projection** ( $\approx 917.83 \approx 918$  on Qwen2.5-7B; stable rank  $\approx 10.04$ , participation ratio  $\approx 93.45$ , mean  $|\cos|$  0.082, p95 0.233) are different quantities, not a contradiction:  $\approx 918$  is a **spectral property of the output projection  $U$  alone** (an entropy-effective rank), *not* an operational capacity — it bounds nothing we prove here; the across-decision capacity is governed by **key-disjointness** (above), not by  $U$ ’s spectrum, while  $\approx 2$  counts how many targets can *co-win within one* decision. Both are empirical observations. The clean derivation of the  $\approx 2$  ceiling from the Gram structure of  $C$  plus softmax saturation, and any exact link to the effective rank, is left for future work. We summarize: *capacity is effectively unbounded across key-disjoint decisions and  $\approx 2$  within a single decision — both empirical observations, not a proved bound.*

## 6 The basis-privilege probes (falsification battery)

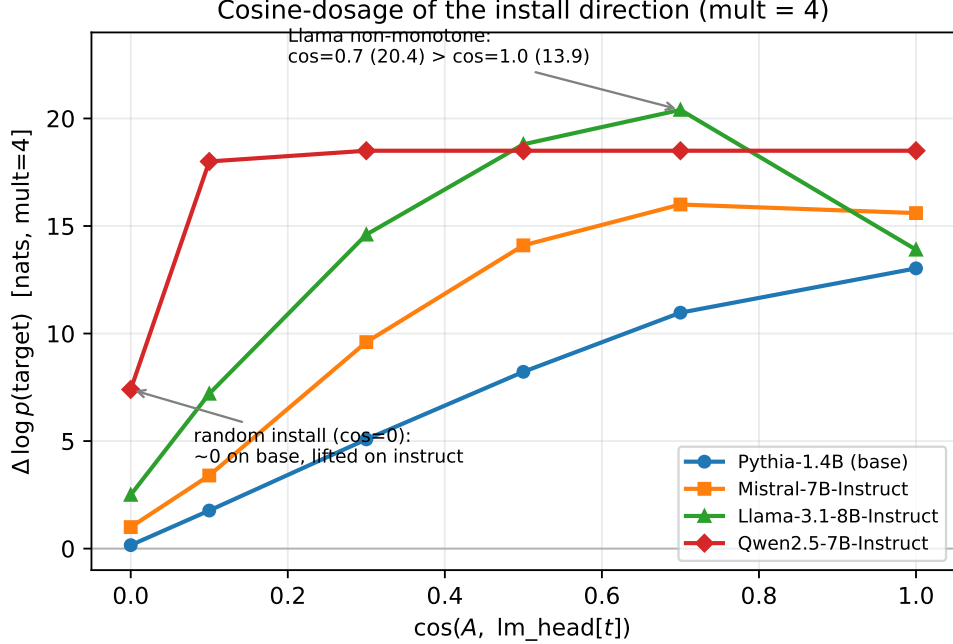
T1 is *exact* because the stored value is the `lm_head` row (under direct addition; §2). Seven falsification probes isolate that the privileged write direction is specifically the **output-projection row of the target token**. Each replaces or perturbs the installed value and asks whether the install still **flips the argmax**; at the flip level, only the genuine `lm_head` row survives (with one important off-axis caveat stated at the end of this section). Figures below are on Qwen2.5-7B unless noted. These probes support the *mechanism* reading of T1 — that the privileged write direction is the unembedding row — and are not required for the T2 bound, which is purely algebraic.

| # | Probe                   | Control it kills                                                            | Measured outcome                                                                                                                                                                                                           |
|---|-------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Row-specific            | random vector @ matched norm                                                | $\approx 0$ recall; only the true row works                                                                                                                                                                                |
| 2 | Subspace                | random vector in top- $k$ right-singular subspace, $k \in \{10, 100, 918\}$ | $\approx 0$ recall, <b>no monotone ramp in <math>k</math></b>                                                                                                                                                              |
| 3 | Token-identity          | <code>lm_head[(t+offset) mod V]</code>                                      | <b>zero</b> canonical-target recall (biases the offset token)                                                                                                                                                              |
| 4 | Output-projection       | <code>embed_tokens[t]</code> (input embedding), untied model                | <b>zero</b> recall — privilege is the <i>output</i> projection                                                                                                                                                             |
| 5 | Layer-stability         | earlier/late layers                                                         | random-A $\approx 0$ at every layer; canonical shows a sharp $\sim 4\times$ single-layer jump at the operating layer                                                                                                       |
| 6 | Magnitude               | row-norm correlation                                                        | recall $\perp$ row norm: Pearson $r \approx 0.094$ , Spearman $\rho \approx 0.167$ (norms span 0.52–0.87)                                                                                                                  |
| 7 | Effective-rank capacity | —                                                                           | mean $ \cos $ 0.082, median 0.061, p95 0.233; stable rank $\approx 10.04$ ; participation ratio $\approx 93.45$ ; <b>entropy effective rank</b> $\approx 917.83$ ( $\approx 918$ ); condition $\# \approx 2.2 \times 10^6$ |

**Cross-family.** The  $0.940 \rightarrow 0.000$  paraphrased-recall collapse under random-A reproduces on Qwen2.5-7B (L27), Llama-3.1-8B (L30), and Mistral-7B-v0.3 (L30); the strong-form privilege is further confirmed on **untied-embedding** Pythia-1.4B and Pythia-12B (canonical +14.07/+14.08 nats, 10/10 flips, vs random-A +0.37/+0.31;  $38\times/45\times$ ), i.e. 5 architectures / 4 families, 2 verifiably untied. The **orthogonal-A control** — projecting the `lm_head[t]` component out and renorming — is a *pre-registered 4-architecture (Pythia-1.4B / Qwen-7B / Llama-8B / Mistral-7B), 5-seed, cosine-dosage falsification*: it collapses the effect to  $\approx$ random-A on **4/4 archs** (orth/canonical ratio: Pythia 1.2%, Mistral 5.8%, Llama 18%, Qwen 38%), evidence that the privilege rests on the unembedding-readout geometry, i.e. it is exactly the T1 mechanism and not a residual-stream-internal computation.

**Off-axis caveat (the honest scope of “only `lm_head` survives”).** The probes establish privilege *at the argmax-flip level*: the orthogonal-A install never flips the target (0/10), and flips track  $\cos(A, \text{lm\_head}[t])$ . They do **not** claim a wrong direction moves the target logit by zero. On instruction-tuned, chat-templated models a **norm-matched random install** lifts the target logit by a non-trivial amount at large scale ( $\approx +7$  nats Qwen, +2.7 Llama, +1 Mistral at  $\times 4$ ;  $\approx 0$  on base Pythia), and the cosine-dosage is **non-monotone on Llama** ( $\cos = 0.7 \rightarrow +20.4$  nats *exceeds* canonical  $\cos = 1.0 \rightarrow +13.9$ ). This is a residual *attractor*, not a basis privilege — it decomposes

as  $\Delta \log p \approx \text{DLA}(\cos) + \text{Attractor}(\sqrt{1 - \cos^2})$ , and the orthogonal-A control rules out a non-DLA mechanism on all 4 archs. We therefore describe  $A = \text{lm\_head}[t]$  as the **maximum-readout-coupling deployment direction** (it owns the argmax flip), **not** as a “discovered mechanism.”



**Figure 2:** *Cosine-dosage of the install direction (norm  $\text{mult} = 4$ ; medians over seeds; Appendix B.9).*  $\Delta \log p(\text{target})$  vs  $\cos(A, \text{lm\_head}[t])$ , with  $A = c \cdot \hat{e}_{\text{canon}} + \sqrt{1 - c^2} \cdot R_{\perp}$ . On the **base** model (Pythia-1.4B) the lift is  $\approx$ linear in  $\cos$  and  $\approx 0$  at  $\cos = 0$  — direct logit attribution. On **instruction-tuned, chat-templated** models a norm-matched *random* install ( $\cos = 0$ ) already lifts the target (+7.4 Qwen / +2.5 Llama / +1.0 Mistral nats) and **Llama is non-monotone** ( $\cos = 0.7 \rightarrow +20.4$  exceeds canonical  $\cos = 1.0 \rightarrow +13.9$ ): an off-axis residual attractor,  $\Delta \log p \approx \text{DLA}(\cos) + \text{Attractor}(\sqrt{1 - \cos^2})$ . The orthogonal-A control ( $\cos = 0$ ) never flips the argmax on any architecture, so the *argmax-flip* privilege still belongs to  $\text{lm\_head}[t]$ . (Values: the §6 cosine-dosage sweep, medians over seeds.)

## 7 The empirical map (split by regime)

We separate evidence at the **final-residual install site** (Tier A — literally eq. (1), the final-norm pre-hook, where T1 is exact) from evidence at a **late-layer install site** (Tier B — the operating layer L27/L30, with downstream nonlinear blocks still to run). The tiers test different things: Tier A tests the exact closed form, while in Tier B the remaining layers *could* in principle rescue input-conditional computation. The honest reading of Tier B is an **empirical pattern**, within one site, consistent with the readout-regime limit (T2 does not formally govern late-layer installs — §10): tasks that need only a **bias** succeed (single-token recall, key-disjoint co-load) while tasks that need **computation or chaining** hit the wall. Mixing the tiers would be a category error; each row states its injection site.

**What is load-bearing.** The two theorems stand *algebraically* — no experiment is required to establish T1, T2, or T2' (Appendix A). The closed-form check ( $\approx 5 \times 10^{-6}$ ) is a sanity test on the algebra, not support for the claim. The corollaries C1–C3 are *supported, not proved*, by the

tiered map below; the capacity figures (§5) are *exploratory* observations. The map exists to show the limit **binds in practice**, not to carry the theorem.

| Claim                                            | Rests on                         | Role of experiment                                       |
|--------------------------------------------------|----------------------------------|----------------------------------------------------------|
| T1 (closed form)                                 | Lemmas 1/3/4 (algebra)           | sanity check only (max-diff $\approx 5 \times 10^{-6}$ ) |
| T2 / T2' (cone-confinement; rank $\leq m$ )      | Lemmas 1–4 (algebra)             | none — proved without experiment                         |
| C1 (no hard override)                            | T2(b)                            | supporting (Tier B readout-wall probe)                   |
| C2 (only tips)                                   | T2                               | supporting (Tier A/B scaffolded composition + controls)  |
| C3 (cannot compute a hidden var)                 | T2/T2' + surface property (§4.3) | supporting (Tier B multi-hop bars)                       |
| Capacity ( $\approx 2$ within / additive across) | —                                | exploratory observation (no proof)                       |

### Tier A — genuine final-residual installs (direct tests of T1/T2)

| Observation                           | Injection site | Decisive numbers                                                                                                                                                                 | Supports                  |
|---------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Closed form <b>is</b> the rank-1 bias | final residual | direct vs predicted max-diff 4.77e-6 (Mistral, $\alpha = 24$ ), 7.15e-6 (Qwen, $\alpha = 64$ ); rank-1 term = median 96.1% (Mistral; 96.83% Qwen) of $\ b\ $ , argmax-preserving | T1, T2(c)                 |
| Behavioral install                    | final residual | cross-arch: Llama +24.61 / Mistral +20.40 (n=20) / Qwen +17.25 (n=5 smoke) nats, 0 leakage                                                                                       | C2 (tips a propensity)    |
| K-target competition                  | final residual | $\geq 1$ -of-2 wins: Llama 14/15, Mistral 15/15, Qwen 11/15 (strict both-win 3/2/1 of 15) vs random control 0/1600; winner-take-all, $K \geq 3$ collapses                        | §5 ( $\approx 2$ ceiling) |

### Tier B — late-layer installs (downstream nonlinearity available; bias-tasks pass, compute-tasks wall)

| Observation                                              | Injection site                            | Decisive numbers                                                                                                                                                                                                                                                                      | Supports / caveat                                               |
|----------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Single-token recall, flat prior ( <b>bias suffices</b> ) | late layer (L27 Qwen / L30 Llama-Mistral) | oracle $\rho = 1.000$ across <b>6 models, 1.5B–72B, 3 families</b> (Qwen 1.5B/7B/14B/72B, Llama-8B, Mistral-7B), full $\rho$ 0.900–0.953; $\approx 13 \times / 47 \times$ over MEMIT/ROME paraphrased, $\geq$ top-3 RAG with 0 retrieved tokens; gap to oracle = routing, not content | late-site <b>positive</b> : only a bias is needed (cross-scale) |

| Observation                                                            | Injection site                                   | Decisive numbers                                                                                                                                                                                                    | Supports / caveat                                                                                                                                                            |
|------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Co-load additivity<br>( <b>key-disjoint</b> )                          | late layer<br>(L27/L30)                          | 43-KB / ~2,580-fact per-<br>KB co-load $\Delta = +0.00$<br>pp @ p95; <b>Appl=0</b> ; bit-<br>identical no-harm (0/15,498<br>spurious activations over<br>MMLU+GSM8K+HumanEval);<br>idempotent                       | late-site <b>positive</b> :<br>§5 additivity, no<br>computation                                                                                                              |
| Composition, <b>visi-<br/>ble</b> operand                              | mid-stack<br>(L18/21)                            | 272/272 at <b>oracle/exact-delta</b><br>(upper bound); learned hidden<br>route 189/272=0.695 <b>fails its<br/>gate</b> (K=4-codebook regression,<br>not recovery); controls no-module<br>0.2684, wrong-delta 0.0551 | C2 (in-context,<br><b>not</b> module<br>recovery)                                                                                                                            |
| Residual injection,<br><b>table visible</b>                            | late residual                                    | 32/32; no-bridge floor 7/32; swap<br>controls follow injected identity                                                                                                                                              | C2 (near-<br>balanced, visibly<br>scaffolded);<br>positive control<br>passes                                                                                                 |
| Any-token install<br>(saturating)                                      | late MLP (L27)                                   | target +20.00 nats (saturated),<br>5/5 flip; random-A -0.44 (45×)                                                                                                                                                   | T1; an unbounded<br>blunt overwrite,<br>not a C2 tip                                                                                                                         |
| Oracle-correct<br>content fails to<br>move a peaked<br>logit (bounded) | bank @ readout                                   | +0.016...+0.136 logit at $\times 1 \dots \times 8$ ;<br>stays 8/32                                                                                                                                                  | C1 (no override<br>guarantee); single-<br>hop, visible bridge                                                                                                                |
| Single-pass multi-<br>hop (Llama-3.1-<br>8B)                           | first-hop install<br>(subject pos)               | 0.183 vs 0.40 bar (n=372,<br>McNemar p=2.15e-20; locus-<br>confounded); Llama→Mistral<br>cross-host 0.120 vs 0.70 (n=50)                                                                                            | C3 (no chaining<br>absent an exposed<br>bridge)                                                                                                                              |
| Mid-layer “read”<br>is bias-equivalent                                 | mid-layer (toy<br>from-scratch<br>CCA substrate) | cosine 0.9918 ( $\geq 0.95$ kill); com-<br>bine 2/16 < 0.25 chance; payload-<br>swap 0/3                                                                                                                            | C3 (read carries<br>no composable<br>content here)                                                                                                                           |
| Hidden second-<br>hop surface                                          | hidden module                                    | ceiling 8/32                                                                                                                                                                                                        | failed<br>construction:<br>positive control<br>fails 8/32, so it<br>cannot distinguish<br>a T2 wall from<br>a broken surface;<br>reported as such,<br><b>not</b> as evidence |

**On the “matched pair.”** It is tempting to present the visible-table 32/32 and the hidden-hop 8/32 as one controlled before/after. They are **two different surfaces**, and the hidden-hop run failed its own positive control — so it cannot by itself separate the T2 wall from a broken surface. The honest exhibits are: **scaffolded success** = the 272/272 composition (oracle/exact-delta upper bound; clean controls) and the visible-table 32/32 (passing positive control); **no override** = the readout-wall diagnostic (oracle-correct content, +0.016... + 0.136 logit, single-hop); **cannot chain** = the single-pass multi-hop bars and the mid-layer-read discriminator. The broken hidden-hop surface is reported as such, not as evidence.

Reframed across the map: the final-residual installs (Tier A) behave exactly as the closed form predicts; and at the late-layer site (Tier B) the same pattern is empirically visible — tasks needing only a bias (recall, key-disjoint co-load) succeed, while tasks needing computation or chaining

(hidden-route composition, single-pass multi-hop) hit the wall.

## 8 Related work

**The established landscape.** Two long-standing lines of work read and write the residual stream. *Reading:* the logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2023, arXiv:2303.08112) decode hidden states through the unembedding; Geva et al. (2021, arXiv:2012.14913; 2022, arXiv:2203.14680) read FFN layers as key-value memories projected into vocabulary space. *Writing:* activation/representation steering — function vectors (Todd et al., 2024, arXiv:2310.15213; Hendel et al., 2023, arXiv:2310.15916) and steering vectors (Turner et al., 2023, arXiv:2308.10248; Zou et al., 2023, arXiv:2310.01405; Panickssery et al. / CAA, 2024, arXiv:2312.06681) — adds directions in the **representation regime** (mid/early layers), while knowledge editors that *write to weights* via gradient-derived rank-one updates (ROME; Meng et al., 2022, arXiv:2202.05262) perform a persistent weight edit, not an inference-time additive install. The representation-engineering survey (Bartoszcze et al., 2025, arXiv:2502.17601) catalogs *cost-type* risks of steering — performance degradation, compute, and steerability — which are resource/quality trade-offs distinct from T2’s *structural* limit on what can be computed at all. Against this landscape, we characterize *writing in the readout regime* (the final residual) and prove its ceiling.

**Closest contemporaneous work (2026).** *Closest mechanism.* Wind (2026), *Prometheus Mind: Retrofitting Memory to Frozen Language Models* (arXiv:2601.15324), §4.4.3 (“Identity V”), already uses a row of `lm_head` as an attention value and states that “row  $i$  of `lm_head` is precisely the direction that produces high probability for token  $i$ ,” injecting  $v_{\text{memory}} = \alpha \cdot W_{lm}[\text{token}] / \|W_{lm}[\text{token}]\| \cdot \|h\|$  into an attention KV pair at one layer for the first generated token. **T1’s core observation is therefore not new.** Wind has **no** impossibility result, no falsification-probe family, and no capacity theory; our contribution is T2 (cone-confinement) + the probes + the capacity observations. (Mechanistically Wind injects through attention — hence through  $W_O$  — vs. our direct final-residual install; that is the mechanism axis, while this paper is about the *theory* of the readout regime.)

*Complementary regime.* Nadaf (2026), *Steerable but Not Decodable: Function Vectors Operate Beyond the Logit Lens* (arXiv:2604.02608), shows **empirically** — across 12 tasks, 6 models, and 4,032 cross-template transfers — that mid-layer function vectors steer behavior *even where the logit lens cannot decode the answer at any layer*, with FVs that reach  $> 0.90$  steering accuracy still projecting to incoherent token distributions (“FVs encode computational instructions rather than answer directions”). That is precisely a re-ranking effect whose residual is **not** a fixed `lm_head`-row bias — the **representation regime**. Nadaf also reports a model-family divergence (Mistral FVs rewrite intermediate representations; Llama/Gemma produce near-zero logit-lens changes despite successful steering), underscoring that the representation regime’s mechanism is *not* the uniform readout-bias our T2 characterizes. We supply the closed-form theory of the **complementary readout regime** and prove its cone-confinement; cited together, the two describe the two-regime picture of §1/§4.4.

*Architecture and empirical axes.* *WriteSAE: Sparse Autoencoders for Recurrent State* (Young, 2026, arXiv:2605.12770) independently finds an install→logit closed form for the **recurrent/state-space write site**: on Gated DeltaNet a formula in the forget gate, read query, and **output embedding** predicts the resulting logit change at  $R^2 = 0.98$  (88.1% transfer to Mamba-2-370M) — the SSM analogue of T1, where the output-embedding row again governs the readout, but with

no temperature, cone, capacity, or falsification-probe results; it neither scoops T1’s transformer-exact form nor T2. Billa (2026, arXiv:2604.15557), the *Linear Accessibility Profile*, is an **empirical** predictor of *where* a difference-of-means steering vector succeeds (it applies the unembedding to hidden states), with a three-regime accessibility taxonomy — complementary to, not overlapping, our *structural* bound. The blog essay *The Mechanism of Logit Gap Steering* (toooold.com, 2026-02-09) treats prompt- and activation-steering as one operation via an “identity-propagator” view of the residual stream — an *approximate* late-layer  $\approx$  logit-bias equivalence using generic difference-of-means vectors, with **no** temperature term, **no** impossibility, and **no** capacity bound. (Distinct from the similarly-named jailbreak-suffix paper *Logit-Gap Steering*, arXiv:2506.24056, which we do not rely on.) We give the *exact* equivalence tied to `lm_head[token]` (which is *why* it is exact), the temperature, the impossibility (T2), and the capacity observations.

## 9 Why it matters: the control reading

T2 is also a control result. A readout install is an **auditable** control: by T1 its effect is a removable, composable bias plus a ranking-preserving temperature — unlike fine-tuning or prompting, you can write down exactly what it does (the closed form matches direct computation to  $\approx 5 \times 10^{-6}$ ) and undo it bit-exactly. (In the no-norm always-on case the bias is literally input-independent; under RMSNorm/LayerNorm with key-addressed attention it is input-modulated through  $b_i(x)$  and  $s(x)$  but stays confined to the fixed cone — still closed-form and removable, just not a single constant.) But T2 says exactly what it *cannot* do, prescribing a two-layer division of labor:

- **Put in the readout regime** the jobs that only need to **tip a propensity**: behavior, tone, persona, refusal/anti-targeting, tool-call suppression. These are near-balanced decisions (C2); a fixed cone bias is the right, auditable instrument, and the final-residual evidence (Tier A: the behavioral install) confirms it works there.
- **Do not put in the readout regime** any job that needs a **hard guarantee** — authorization gates, capability locks, content that must *override* a determined prior or depend on a hidden computed value. By C1/C3 a bounded readout bias offers no override guarantee against a peaked prior and cannot carry a hidden intermediate. Such guarantees require a **deterministic override outside the model** (a gate that does not depend on the model winning a logit race).  
Tip in the readout regime; gate deterministically outside it.

## 10 Limitations and open work

- **Capacity derivation.** The single-decision  $\approx 2$  ceiling is empirical; deriving it from the Gram structure of  $C$  and softmax saturation, and relating it exactly to the effective rank ( $\approx 918$ ), is open.
- **Late-stack  $\leftrightarrow$  final-residual.** The closed form bridges a *single* late install to the final-residual object; the multi-slot late-stack case (Tier B) is empirically consistent but not proven to collapse to eq. (1). Either restrict load-bearing claims to final-residual installs (Tier A) or prove the reduction.
- **The regime distinction is not a lattice.** We prove readout cone-confinement and cite empirical representation-regime computation; a model-independent formal separation (with both a concrete witness and the inclusion direction) is not established and is left open.
- $W_O$  / **attention-addressed memories.** Lemma 1’s exact Gram-column form and the §6 probes assume direct residual addition; the attention-head version stays cone-confined (T2(c))



but its exact bias direction is  $U \cdot W_O \cdot U[t]$ . A clean treatment of that case would broaden the probe story.

- **Breadth.** T1/T2 are architecture-general (final-residual + unembedding only); the empirical map is Qwen-centric with cross-family probe support. Replicating the readout-wall (C1) and the single-pass-chain limit (C3) end-to-end on  $\geq 2$  more families is high-value.
- **Reproduction.** Forward-only instrumentation on public checkpoints; every parameter needed to re-derive each reported number is specified in §3, §6, and Appendix B (see *Reproducibility*, below).

## 11 Conclusion

A final-residual additive intervention on a frozen transformer — of which installing a scaled `lm_head` row is the exactly-characterizable case — acts on every input as  $\tilde{z}(x) = s(x) \cdot z(x) + c(x)$ : a positive scalar temperature plus a re-ranking term drawn from a single fixed, low-dimensional set of directions chosen *before any input* (T1, verified to  $\approx 5 \times 10^{-6}$ ). The content is not the equivalence — which is known (Wind, 2026) — but its **structure**: because the re-ranking directions are fixed before the input and the addressing weights are non-negative, no input can produce a re-ranking direction outside the pre-installed set, however nonlinear the addressor (T2; the measurable form is the affine-rank bound T2'). This converts a body of empirical phenomena into corollaries — a bounded install cannot *override* a peaked prior, can only *tip* a context-scaffolded near-balanced decision, and cannot *compute or carry* a hidden intermediate the addressor does not already expose — and the empirical map shows the limit binds in practice: final-residual installs (Tier A) behave exactly as the closed form predicts, and at the late-layer site (Tier B) the same pattern is empirically visible — bias-only tasks succeed while computation and chaining hit the wall. The control reading is the payoff: the readout regime is the right, auditable place for jobs that only need to *tip* a propensity, and the wrong place for any hard guarantee, which requires a deterministic override outside the model. The boundary, stated precisely, is the contribution.

## Reproducibility

Every result uses publicly available instruction-tuned checkpoints (Qwen2.5- $\{1.5\text{B}, 7\text{B}, 14\text{B}, 72\text{B}\}$ , Llama-3.1-8B, Mistral-7B-v0.3, Pythia-1.4B/12B) and forward-only instrumentation — standard forward hooks; no weights are trained or modified. The paper specifies, for each result, everything needed to rebuild it: the exact checkpoint; the install site (final-norm pre-hook vs. late-MLP post-hook) and operating layer; the stored value  $\alpha \cdot U[t]$  and the relative-norm calibration that sets  $\alpha$ ; the metric; the decoding setting (greedy/argmax at the install position); and the seeds (Appendix B). T1's closed form is a two-forward-pass identity (B.4); the basis-privilege probes are direct value substitutions (B.9); the capacity and corollary probes are specified in B.6–B.8; the empirical datasets are named where used (MMLU, GSM8K, HumanEval for the no-harm sweep; MQuAKE-CF for the multi-hop chains). Because the construction is forward-only and the parameters are given, each reported number is re-derivable from the public checkpoints and the procedures in §3, §6, and Appendix B.

A companion repository — <https://github.com/orbitnate/readout-regime> (MIT) — provides the authentic script and the committed result artifact behind every reported number, plus a model-free check that reproduces the T1 identity in seconds (no model or GPU). The closed-form verification (§3), the `lm_head` effective-rank measurement (§6), and the K-target capacity probe

(§5) are clone-and-run on a single 7B checkpoint; the larger cross-scale recall and the Tier-B results ship as the original run artifacts with their source scripts. The MIT license covers the code only and grants no patent rights.

## Broader impact

This is a theoretical and diagnostic contribution about what a frozen model’s readout-space controls can and cannot do; it introduces no new capability and uses only public models. Its main practical consequence is safety-positive: the impossibility (T2) tells builders *not* to rely on a readout-space bias for any hard guarantee — authorization, capability locks, content that must override a determined prior — and to place such guarantees in a deterministic mechanism outside the model, reserving the readout regime for auditable, removable propensity tips. We note the usual dual-use caveat that any characterization of steering can inform both helpful and harmful steering; the result here narrows rather than expands what residual-space steering can achieve.

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## A Proof details

**A.1 Lemma 1.**  $U(h + \Delta) = z + U\Delta$ ,  $U\Delta = \sum a_i \alpha_i U U[t_i] = \sum a_i \alpha_i g_{t_i}$ .  $\square$

**A.2 Lemma 2.**  $g_{t_i} = U U[t_i] \in \text{col}(U)$ . The set  $\{\sum_i a_i \alpha_i g_{t_i} : a_i \geq 0\}$  is the convex cone over the **signed installed generators**  $\{\alpha_i g_{t_i}\}$ : if all  $\alpha_i \geq 0$  it is cone $\{g_{t_i}\}$ ; if some  $\alpha_i < 0$  it is the cone over those signed generators, whose linear span is contained in  $\text{span}\{g_{t_i}\}$  (dimension  $\leq m$ ) and equals the full subspace  $\text{span}\{g_{t_i}\}$  **only if** the installed set supplies enough opposite-signed directions. In all cases  $\dim \text{span} \leq m$ . All of  $(U, \{t_i\}, \{\alpha_i\})$  are fixed before input.  $\square$

**A.3 T2(a)/(b).** No-norm always-on:  $\tilde{z} = z + b$ ,  $b = \sum \alpha_i g_{t_i}$  fixed;  $\text{softmax}(z + b)_j = p_j e^{b_j} / \sum_k p_k e^{b_k}$  depends on  $x$  only through  $p(x)$ ; equal  $p \Rightarrow$  equal output. Gap  $\tilde{z}_j - \tilde{z}_k = s(x)(z_j - z_k) + (c_j(x) - c_k(x))$ ,  $c(x) \in C$  (Lemmas 1/3/4);  $s \equiv 1$ ,  $c$  constant in the no-norm always-on case.  $\square$

**A.4 T2(c) and T2'.** By Lemmas 3/4,  $\tilde{z}(x) = s(x)z(x) + c(x)$ ,  $c(x) = \sum_i b_i(x)u_{t_i} \in C$ ,  $\dim \text{span}(C) \leq m$ ,  $C$  fixed before input;  $s(x) > 0$  preserves the ranking of  $z(x)$ ; an attention sink gives  $\sum a_i \leq 1$ , keeping  $c(x) \in C$ . Hence  $d(x) := \tilde{z}(x) - s(x)z(x) = c(x) \in C$  for all  $x$ , so the affine dimension of  $\{d(x)\}$  is  $\leq \dim \text{span}(C) \leq m$ .  $\square$

(No  $\square$  is claimed for any  $G_{\text{read}} \subseteq G_{\text{repr}} / \subsetneq$  statement; per §4.4 the lattice is not asserted.)

## B Experimental details

All experiments use publicly available instruction-tuned checkpoints and standard forward-pass instrumentation (forward hooks); no weights are trained or modified. We describe each measurement in enough detail to reproduce it from the public model alone.

**B.1 Models.** Qwen2.5-7B-Instruct (28 layers, hidden 3584, vocab 152064, RMSNorm, `tie_word_embeddings=False`); Llama-3.1-8B-Instruct; Mistral-7B-Instruct-v0.3; Pythia-1.4B and Pythia-12B (both with untied input/output embeddings). All use a final RMSNorm followed by the unembedding `lm_head`, except where the LayerNorm analysis of Lemma 4 is invoked.

**B.2 Install sites.** Two sites are used and always reported per result. (i) *Final residual*: a pre-hook on the final normalization layer adds the install vector to the last-position residual immediately before normalization, so it reaches `lm_head` through the norm only — this is the direct-addition object of eq. (1) (Tier A). (ii) *Late MLP down-projection*: a post-hook on a selected layer’s MLP down-projection adds the install to that layer’s output, after which the remaining blocks run normally (Tier B). The operating layer for (ii) is the layer maximizing the basis-privilege gap of §6 (e.g. L27 on Qwen, L30 on Llama/Mistral).

**B.3 Install value and scale.** The stored value is  $\alpha \cdot U[t]$ , the scaled `lm_head` row of target

token  $t$ ; for the K-target and edit constructions it is a sum/difference of such rows.  $\alpha$  is set by a relative-norm calibration (so the added vector’s norm is a fixed fraction of the host residual norm at the install position); the per-model  $\alpha$  operating points reported in §7 are stated inline with each result (e.g. the any-token and behavioral installs use the saturating regime).

**B.4 Closed-form verification (Lemma 3, T1).** For random prompts, we collect the last-position final-layer residual  $h$ , compute the predicted bias  $b_{\text{pred}} = (\alpha/r') \cdot u_t + ((r-r')/(r \cdot r')) \cdot U(\gamma \odot h)$  (with  $r, r'$  the RMSNorm denominators before/after install,  $u_t = U(\gamma \odot U[t])$ ), and compare to the *direct* bias  $b_{\text{dir}} = \text{logits}(h + \alpha U[t]) - \text{logits}(h)$  obtained by two forward evaluations. We report  $\max |b_{\text{pred}} - b_{\text{dir}}|$  over the vocabulary and prompts ( $4.77 \times 10^{-6}$  Mistral at  $\alpha = 24$ ,  $7.15 \times 10^{-6}$  Qwen at  $\alpha = 64$ ), the fraction of  $\|b_{\text{dir}}\|$  carried by the rank-1 term (median 96.1% on Mistral, 96.83% on Qwen), and that the rescale term leaves the argmax unchanged.

**B.5 A falsification recipe for “is this intervention a readout install?” (T2’).** Corollary T2’ is a *proven* bound (Appendix A.4): a readout install’s re-ranking residual at the structural temperature has affine rank  $\leq m$  by construction, so measuring *that* is tautological and we do not report it as evidence. The useful, falsifiable test runs the other way — on an intervention of *unknown* type: record the installed logits  $\tilde{z}(x)$  and base logits  $z(x)$ , form  $d(x) = \tilde{z}(x) - s(x)z(x)$  at the **structural** temperature  $s(x) = r(h(x))/r(\tilde{h}(x))$  (Lemma 3 — **not** a least-squares fit, which would absorb part of  $c(x)$  and inflate the rank), and report the affine rank of  $\{d(x)\}$ . An intervention whose residual rank exceeds its slot count  $m$  is **not** realizable as an  $m$ -slot readout install. The natural application — measuring a mid-layer representation intervention’s residual rank to confirm it *exceeds* the readout bound, turning the two-regime separation of §4.4 from cited into measured — is left to future work.

**B.6 Bounded-override probe (C1).** With the install direction set to the oracle-correct `lm_head[target]` (cosine  $\approx 1.0$  with the target row), we sweep the scale ( $\times 1 \dots \times 8$ ) at the final-residual install and report the change in the target-token logit and whether the argmax flips; under bounded calibration the target logit moves by only  $+0.016 \dots +0.136$  and the peaked prior does not flip.

**B.7 K-target competition (§5).** We co-install  $K$  target rows at one final-residual position and report, per architecture, the  $\geq 1$ -of-2-wins rate and the **strict both-win** rate at  $K=2$  (and the  $K \geq 3$  collapse), each against a norm-matched random-row control (zero hits across 1,600 random control packets). The readout is winner-take-all, so co-installed rows compete rather than add.

**B.8 Co-load additivity (§5).** We load up to 43 single-fact modules ( $\approx 2,580$  facts) addressed by disjoint keys, and report the per-module recall solo vs. co-loaded ( $\Delta$  at the 95th percentile), the perplexity change against the unmodified model on held-out text (App1), and idempotence of load/unload.

**B.9 Basis-privilege probes (§6).** Each probe replaces the install value with a control (random vector at matched norm; random vector in the top- $k$  right-singular subspace; wrong-token row; input embedding row; row at a different layer; row scaled across the norm range) and reports recall vs. the genuine `lm_head` row. The effective-rank measurement (probe 7) samples vocabulary rows, forms the row matrix, and reports pairwise  $|\cos|$  statistics and the singular-spectrum measures (stable rank, participation ratio, entropy effective rank) of the output projection.

**B.10 Decoding and seeds.** Unless a probe specifies otherwise, scoring is greedy/argmax at the install position; the readout-modifier sensitivity of T1 (argmax-preserving vs. tail-sensitive readouts) is reported separately. Probes that draw random prompts or random control directions fix a seed per run; the reported figures are stable across the seeds used.